定态微扰

定态微扰理论

目的：

求哈密顿算符本征值方程 $\hat{H}\psi_n = E_n\psi_n$ 近似解 ψ_n 与 E_n

假设：

$\hat{H}$ 不显含时间，可分为可求解的 $\hat{H}^{(0)}$ 与微扰 $\hat{H}'$ 两部份。

$$\hat{H} = \hat{H}^{(0)} + \hat{H}'$$

$$\hat{H}^{(0)}\psi_0^{(0)} = E_n^{(0)}\psi_n^{(0)}$$

微扰参数

$$\hat{H}' \equiv \lambda\hat{H}^{(1)}$$

$$E_n = E_n^{(0)} + E_n^{(1)} + E_n^{(2)} + \cdots$$

$$= E_n^{(0)} + \lambda e_n^{(1)} + \lambda^2 e_n^{(2)} + \cdots$$

表征微扰程度。

$$\psi_n = \psi_n^{(0)} + \lambda\phi_n^{(1)} + \lambda^2\phi_n^{(2)} + \cdots$$

$$= \psi_n^{(0)} + \psi_n^{(1)} + \psi_n^{(2)} + \cdots$$

$E_n^{(0)}$、$\psi_n^{(0)}$ 表体系未受微扰时能量与波函数，称零级近似能量/波函数。

本征值与本征函数的修正方程

$$\hat{H}\psi_n = E_n\psi_n \begin{cases} \hat{H} = \hat{H}^{(0)} + \lambda\hat{H}^{(1)} = \hat{H}^{(0)} + \hat{H}' \\ E_n = E_n^{(0)} + \lambda e_n^{(1)} + \lambda^2 e_n^{(2)} = E_n^{(0)} + E_n^{(1)} + E_n^{(2)} \\ \psi_n = \psi_n^{(0)} + \lambda\phi_n^{(1)} + \lambda^2\phi_n^{(2)} = \psi_n^{(0)} + \phi_n^{(1)} + \phi_n^{(2)} \end{cases}$$

$$\text{左边} = \hat{H}\psi_n = (\hat{H}^{(0)} + \lambda\hat{H}^{(1)})(\psi_n^{(0)} + \lambda\phi_n^{(1)} + \lambda^2\phi_n^{(2)})$$

$$= \hat{H}^{(0)}\psi_n^{(0)} + \lambda\hat{H}^{(0)}\phi_n^{(1)} + \lambda^2\hat{H}^{(0)}\phi_n^{(2)} + \lambda\hat{H}^{(1)}\phi_n^{(1)} + \lambda^2\hat{H}^{(1)}\phi_n^{(1)} + \lambda^3\hat{H}^{(1)}\phi_n^{(2)}$$

$$\text{右边} = E_n^{(0)}\psi_n^{(0)} + \lambda E_n^{(0)}\phi_n^{(1)} + \lambda^2 E_n^{(0)}\phi_n^{(2)} + \lambda e_n^{(1)}\psi_n^{(0)} + \lambda^2 e_n^{(1)}\phi_n^{(1)} + \lambda^3 e_n^{(1)}\phi_n^{(2)}$$
$$+ \lambda^2 e_n^{(2)}\psi_n^{(0)} + \lambda^3 e_n^{(2)}\phi_n^{(1)} + \lambda^4 e_n^{(2)}\phi_n^{(2)}$$

① $\hat{H}^{(0)}\psi_n^{(0)} = E_n^{(0)}\psi_n^{(0)} \Longrightarrow (\hat{H}^{(0)} - E_n^{(0)})\psi_n^{(0)} = 0$

② $\lambda\hat{H}^{(0)}\phi_n^{(1)} + \lambda\hat{H}^{(1)}\phi_n^{(1)} = \lambda E_n^{(0)}\phi_n^{(1)} + \lambda e_n^{(1)}\psi_n^{(0)}$

$$\Longrightarrow (\hat{H}^{(0)} - E_n^{(0)})\psi_n^{(1)} = -(\hat{H}' - E_n^{(1)})\psi_n^{(0)}$$

③ $\lambda^2\hat{H}^{(0)}\phi_n^{(2)} + \lambda^2\hat{H}^{(1)}\phi_n^{(1)} = \lambda^2 E_n^{(0)}\phi_n^{(2)} + \lambda^2 e_n^{(1)}\phi_n^{(1)} + \lambda^2 e_n^{(2)}\psi_n^{(0)}$

$$\Longrightarrow \hat{H}^{(0)}\psi_n^{(2)} + \hat{H}'\psi_n^{(1)} = E_n^{(0)}\psi_n^{(2)} + E_n^{(1)}\psi_n^{(1)} + E_n^{(2)}\psi_n^{(0)}$$

综上，有本征值与本征函数的修正方程.

$$
\begin{cases}
(\hat{H}^{(0)} - \hat{E}_n^{(0)})\,\psi_n^{(0)} = 0 \\
(\hat{H}^{(0)} - \hat{E}_n^{(0)})\,\psi_n^{(1)} = -(\hat{H}' - \hat{E}_n^{(1)})\,\psi_n^{(0)} \\
(\hat{H}_n^{(0)} - \hat{E}_n^{(0)})\,\psi_n^{(2)} = -(\hat{H}' - \hat{E}_n^{(1)})\,\psi_n^{(1)} + \hat{E}_n^{(2)}\psi_n^{(0)}
\end{cases}
$$

定理

设 $\psi_n^{(1)}$ 为（第二条）方程的解，那么 $\psi_n^{(1)} + a\psi_n^{(0)}$ 也是解.

证明：

设：

$$(\hat{H}^{(0)} - \hat{E}_n^{(0)})\psi_n^{(0)} = 0$$

$$(\hat{H}^{(0)} - \hat{E}_n^{(0)})\psi_n^{(1)} = -(\hat{H}' - E_n^{(1)})\psi_n^{(0)}$$

$\Rightarrow$

$$(\hat{H}^{(0)} - \hat{E}_n^{(0)})(\psi_n^{(1)} + a\psi_n^{(0)}) = -(\hat{H}' - E_n^{(1)})\psi_n^{(0)}$$

其中 $a\psi_n^{(0)} = 0$.

两种微扰理论

① 非简并：属于 $E_n^{(0)}$ 的本征函数仅有1个 $\psi_n^{(0)}$，即 ψ_n 的零级近似

② 简并：属于 $E_n^{(0)}$ 的本征函数有多个 $\psi_{ni}^{(0)}$，需重新组合出 ψ_n 的零级近似

非简并定态微扰

能量一级修正

$$(\hat{H}^{(0)} - E_n^{(0)})\psi_n^{(1)} = (E_n^{(1)} - \hat{H}')\psi_n^{(0)}, \quad (\hat{H}^{(0)} - E_n^{(0)})\psi_n^{(0)} = 0$$

$$\downarrow$$

$$\int \psi_n^{(0)*}(\hat{H}^{(0)} - E_n^{(0)})\psi_n^{(1)} d\tau = \int \psi_n^{(0)*}(E_n^{(1)} - \hat{H}')\psi_n^{(0)} d\tau$$

$$left = \int [(\hat{H}^{(0)} - E_n^{(0)})\psi_n^{(0)}]^* \psi_n^{(1)} d\tau = 0$$

厄米算符

$$\hat{H}^{(0)}\psi_n^{(0)} = E_n^{(0)}\psi_n^{(0)}$$

$$right = \int \psi_n^{(0)*} E_n^{(1)} \psi_n^{(0)} d\tau - \int \psi_n^{(0)*} \hat{H}' \psi_n^{(0)} d\tau = E_n^{(1)} - H'_{nn}$$

所以

$$E_n^{(1)} = H'_{nn}$$

波函数一级修正

已有：

$$(E_n^{(1)} - \hat{H}')\psi_n^{(0)} = (\hat{H}^{(0)} - E_n^{(0)})\psi_n^{(1)}$$

$$(\hat{H}^{(0)} - E_n^{(0)})\psi_n^{(0)} = 0$$

我们将 $\psi_n^{(1)}$ 按 $\hat{H}^{(0)}$ 的本征函数系展开

$$\psi_n^{(1)} = \sum_l a_l^{(1)} \psi_l^{(0)}$$

由于 $\psi_n^{(1)} + a\psi_n^{(0)}$ 仍为方程的解.

$$\psi_n^{(1)} = \sum_l a_l^{(1)} \psi_l^{(0)} - a_l^{(1)} \psi_n^{(0)} = \sum_{l\neq n} a_l^{(1)} \psi_l^{(0)}$$

代入，有：

$$(E_n^{(1)} - \hat{H}')\psi_n^{(0)} = (\hat{H}^{(0)} - E_n^{(0)})\psi_n^{(1)} = (\hat{H}^{(0)} - E_n^{(0)})\sum_{l\neq n} a_l^{(1)} \psi_l^{(0)}$$

$$= \sum_{l\neq n} a_l^{(1)} (E_l^{(0)} - E_n^{(0)})\psi_l^{(0)}$$

两边 $\times \psi_m^{(0)*}$ 积分：

$$\int \psi_m^{(0)*}(E_n^{(1)} - \hat{H}')\psi_n^{(0)} d\tau = \int \psi_m^{(0)*} \sum_{l\neq n} a_l^{(1)} (E_l^{(0)} - E_n^{(0)})\psi_l^{(0)} d\tau \qquad m\neq n$$

$$left = \int [(E_n^{(1)} - \hat{H}')\psi_m^{(0)}]^* \psi_n^{(0)} d\tau = E_n \int \psi_m^{(0)*}\psi_n^{(0)} d\tau - H'_{mn} = -H'_{mn}$$

$$right = \sum_{l\neq n} a_l^{(1)} (E_l^{(0)} - E_n^{(0)}) \int \psi_m^{(0)*}\psi_l^{(0)} d\tau = \sum_{l\neq n} a_l^{(1)} (E_l^{(0)} - E_n^{(0)})\, \delta_{ml}$$

$$= a_m^{(1)} (E_m^{(0)} - E_n^{(0)})$$

由上：

$$a_m^{(1)} = \frac{H'_{mn}}{E_n^{(0)} - E_m^{(0)}} \qquad \psi_n^{(1)} = \sum_{m\neq n} \frac{H'_{mn}}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)}$$

能量二级修正

已知：

$$E_n^{(1)} = H'_{nn} \qquad \psi_n^{(1)} = \sum_{m\neq n} \frac{H'_{mn}}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)} \qquad H'_{mn} = \int \psi_m^{(0)*} H' \psi_n^{(0)} d\tau$$

$$(\hat{H}^{(0)} - E_n^{(0)}) \psi_n^{(2)} = -(\hat{H}' - E_n^{(1)}) \psi_n^{(1)} + E_n^{(2)} \psi_n^{(0)}$$

同上积分

$$(\hat{H}^{(0)} - E_n^{(0)}) \psi_n^{(0)} = 0$$

$$\int \psi_n^{(0)*} (\hat{H}^{(0)} - E_n^{(0)}) \psi_n^{(2)} d\tau = \int \psi_n^{(0)*} [-(\hat{H}' - E_n^{(1)}) \psi_n^{(1)} + E_n^{(2)} \psi_n^{(0)}] d\tau$$

$$Left = \int [(\hat{H}^{(0)} - E_n^{(0)}) \psi_n^{(0)}]^* \psi_n^{(2)} d\tau = 0$$

$$Right = \int \psi_n^{(0)*} E_n^{(2)} \psi_n^{(0)} d\tau - \int \psi_n^{(0)*} \hat{H}' \psi_n^{(1)} d\tau + \int \psi_n^{(0)*} E_n^{(1)} \psi_n^{(1)} d\tau$$

$$= E_n^{(2)} - \int \psi_n^{(0)*} \hat{H}' \sum_{m\neq n} \frac{H'_{mn}}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)} d\tau + \int \psi_n^{(0)*} E_n^{(1)} \sum_{m\neq n} \frac{H'_{mn}}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)} d\tau$$

$$= E_n^{(2)} - \sum_{m\neq n} \frac{H'_{mn}}{E_n^{(0)} - E_m^{(0)}} \left(\underbrace{\int \psi_n^{(0)*} \hat{H}' \psi_m^{(0)} d\tau}_{H'_{mn}} - \underbrace{\int \psi_n^{(0)*} E_n^{(1)} \psi_m^{(0)} d\tau}_{0} \right)$$

$$= E_n^{(2)} - \sum_{m\neq n} \frac{|H'_{mn}|^2}{E_n^{(0)} - E_m^{(0)}}$$

由上，

$$E_n^{(2)} = \sum_{m\neq n} \frac{|H'_{mn}|^2}{E_n^{(0)} - E_m^{(0)}}$$

修正后的能量与波函数

$$E_n = E_n^{(0)} + E_n^{(1)} + E_n^{(2)} + \cdots = E_n^{(0)} + H'_{mn} + \sum_{m\neq n} \frac{H'_{mn}}{E_n^{(0)} - E_m^{(0)}}$$

$$\psi_n = \psi_n^{(0)} + \psi_n^{(1)} + \cdots = \psi_n^{(0)} + \sum_{m\neq n} \frac{H'_{mn}}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)} + \cdots$$

一级微扰适用的条件：

$$\left| \frac{H'_{mn}}{E_n^{(0)} - E_m^{(0)}} \right| << 1 \quad , \quad m \neq n$$

例：电荷为 e_s 的线性谐振子受沿 x 方向恒定弱电场 $\mathcal{E}$ 的作用

1. 哈密顿量

$$\hat{H} = \underbrace{-\frac{\hbar}{2\mu}\frac{d^2}{dx^2} + \frac{1}{2}\mu\omega^2 x^2}_{\hat{H}^{(0)}} \underbrace{- e_s \mathcal{E} x}_{\hat{H}'}$$

2. 能量的一级修正

$$E_n^{(0)} = (n+\frac{1}{2})\hbar\omega \qquad \psi_n^{(0)} = N_n e^{-\alpha^2 x^2/2} H_n(\alpha x)$$

$$E_n^{(1)} = \int \psi_n^{(0)*} \hat{H}' \psi_n^{(0)} d\tau = -\int \psi_n^{(0)*} e_s \mathcal{E} x \psi_n^{(0)} d\tau$$

$$= -e_s \mathcal{E} \int x (\psi_n^{(0)})^2 d\tau = 0$$

3. 能量的 2 级修正

$$x = \frac{1}{2}\left(\frac{2\hbar}{\mu\omega}\right)^{\frac{1}{2}}(\hat{a}+\hat{a}^+)$$

$$\hat{H}' = -e_s \mathcal{E} x$$

$$H'_{mn} = \int \psi_m^{(0)*} \hat{H}' \psi_n^{(0)} d\tau = -e_s \mathcal{E} \int \psi_m^{(0)} x \psi_n^{(0)} d\tau$$

$$= -\sqrt{\frac{\hbar e_s^2 \mathcal{E}^2}{2\mu\omega}} \int \psi_m^{(0)} (\hat{a}+\hat{a}^+) \psi_n^{(0)} d\tau$$

$$= -\sqrt{\frac{\hbar e_s^2 \mathcal{E}^2}{2\mu\omega}} \int \psi_m^{(0)} (\hat{a}\psi_n^{(0)} + \hat{a}^+ \psi_n^{(0)}) d\tau$$

$$= -\sqrt{\frac{\hbar e_s^2 \mathcal{E}^2}{2\mu\omega}} \int \psi_m^{(0)} (\sqrt{n}\,\psi_{n-1}^{(0)} + \sqrt{n+1}\,\psi_{n+1}^{(0)}) d\tau$$

$$= -\sqrt{\frac{\hbar e_s^2 \mathcal{E}^2}{2\mu\omega}} (\sqrt{n}\,\delta_{m,n-1} + \sqrt{n+1}\,\delta_{m,n+1})$$

4. 波函数一级修正

$$H'_{mn} = -\sqrt{\frac{\hbar e_s^2 \mathcal{E}^2}{2\mu\omega}}\left(\sqrt{n}\,\delta_{m,n-1} + \sqrt{n+1}\,\delta_{m,n+1}\right)$$

$$\psi_n^{(1)} = \sum_{m\neq n} \frac{H'_{mn}}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)} = \sum_{m\neq n} \frac{-\sqrt{\frac{\hbar e_s^2 \mathcal{E}^2}{2\mu\omega}}\left(\sqrt{n}\,\delta_{m,n-1} + \sqrt{n+1}\,\delta_{m,n+1}\right)}{(n-m)\hbar\omega} \psi_m^{(0)}$$

$$E_n^{(0)} = \left(n+\frac{1}{2}\right)\hbar\omega$$

$$= -\sqrt{\frac{\hbar e_s^2 \mathcal{E}^2}{2\mu\omega}} \sum_{m\neq n} \frac{\sqrt{n}\,\delta_{m,n-1}\,\psi_m^{(0)}}{(n-m)\hbar\omega} - \sqrt{\frac{\hbar e_s^2 \mathcal{E}^2}{2\mu\omega}} \sum_{m\neq n} \frac{\sqrt{n+1}\,\delta_{m,n+1}\,\psi_m^{(0)}}{(n-m)\hbar\omega}$$

($m=n-1$，$(n-m)=1$；$m=n+1$，$(n-m)=-1$)

$$= -\sqrt{\frac{\hbar e_s^2 \mathcal{E}^2}{2\mu\omega}}\left(-\frac{\sqrt{n}}{\hbar\omega}\psi_{n-1}^{(0)} + \frac{\sqrt{n+1}}{\hbar\omega}\psi_{n+1}^{(0)}\right)$$

$$= \sqrt{\frac{e_s^2 \mathcal{E}^2}{2\mu\hbar\omega^3}}\left(\sqrt{n+1}\,\psi_{n+1}^{(0)} - \sqrt{n}\,\psi_{n-1}^{(0)}\right)$$

说明：

(1) 弱电场使能量下移 $\frac{e_s^2\mathcal{E}^2}{2\mu\omega^2}$　　(2) 体系仍为线性谐振子.

另解：

$$\hat{H} = -\frac{\hbar^2}{2\mu}\frac{d^2}{dx^2} + \frac{1}{2}\mu\omega^2 x^2 - e_s\mathcal{E}x = -\frac{\hbar}{2\mu}\frac{d^2}{dx^2} + \frac{1}{2}\mu^2\omega^2\left[x^2 - \frac{2e_s\mathcal{E}}{\mu\omega^2}x + \left(\frac{e_s\mathcal{E}}{\mu\omega}\right)^2 - \left(\frac{e_s\mathcal{E}}{\mu\omega}\right)^2\right]$$

$$= -\frac{\hbar}{2\mu}\frac{d^2}{dx^2} + \frac{1}{2}\mu^2\omega^2\left(x - \frac{e_s\mathcal{E}}{\mu\omega^2}\right)^2 - \frac{1}{2}\mu\omega^2\left(\frac{e_s\mathcal{E}}{\mu\omega^2}\right)^2 \qquad \text{令 } x' = x - \frac{e_s\mathcal{E}}{\mu\omega^2}$$

$$= -\frac{\hbar}{2\mu}\frac{d^2}{dx^2} + \frac{1}{2}\mu^2\omega^2 x'^2 - \frac{e_s^2\mathcal{E}^2}{2\mu\omega^2}$$

由上明显看到：

① 体系仍为线性谐振子　② 弱电场使能量下移 $\frac{e_s^2\mathcal{E}^2}{2\mu\omega^2}$

③ 平衡点右移 $\frac{e_s\mathcal{E}}{\mu\omega^2}$

简并定态微扰

零级波函数

$$\psi_n^{(0)} = \sum_i c_i^{(0)} \phi_{ni}$$

$$\hat{H}^{(0)} \phi_{ni} = E_n^{(0)} \phi_{ni}$$

$$i = 1, 2, \cdots, k$$

能量的一级修正

$$(\hat{H}^{(0)} - E_n^{(0)}) \psi_n^{(1)} = (E_n^{(1)} - \hat{H}') \psi_n^{(0)} = (E_n^{(1)} - \hat{H}') \sum_i c_i^{(0)} \phi_{ni}$$

$$\int \phi_{nl}^* (\hat{H}^{(0)} - \hat{E}_n^{(0)}) \psi_n^{(1)} d\tau = \int \phi_{nl}^* (E_n^{(1)} - \hat{H}') \sum_i c_i^{(0)} \phi_{ni} d\tau$$

$$Left = \int [(\hat{H}^{(0)} - \hat{E}_n^{(0)}) \phi_{nl}]^* \psi_n^{(1)} d\tau = 0$$

($(\hat{H}^{(0)} - \hat{E}_n^{(0)}) \phi_{nl} = 0$)

$$Right = \sum_i \int \phi_{nl}^* (E_n^{(1)} - \hat{H}') \phi_{ni} d\tau c_i^{(0)} = \sum_i [E_n^{(1)} \int \phi_{nl}^* \phi_{ni} d\tau - \int \phi_{nl}^* \hat{H}' \phi_{ni} d\tau] c_i^{(0)}$$

$$= \sum_i (E_n^{(1)} \delta_{l,i} - H'_{li}) c_i^{(0)}$$

综上，

$$\sum_i (E_n^{(1)} \delta_{l,i} - H'_{li}) c_i^{(0)} = 0$$

其中，

$$H'_{li} = \int \phi_{nl}^* \hat{H}' \phi_{ni} d\tau$$

久期方程

$$0 = \sum_i (H'_{ki} - E_n^{(1)}\delta_{ki})\, c_i^{(0)}$$

$$0 = (H' - E_n^{(1)} I)\, c^{(0)}, \quad c^{(0)} \neq 0$$

$$0 = det\,(H' - E_n^{(1)} I)$$

$$= \begin{vmatrix} H'_{11} - E_n^{(1)} & H'_{12} & \cdots & H'_{1k} \\ H_{21} & H'_{22} - E_n^{(1)} & \cdots & H'_{2k} \\ \cdots & \cdots & \cdots & \cdots \\ H'_{k1} & H'_{k2} & \cdots & H'_{kk} - E_n^{(1)} \end{vmatrix}$$

说明：

- 解方程可得 k 个 $E_n^{(1)}$（记 $E_{nj}^{(1)}$）。$\because E_n = E_n^{(0)} + E_{nj}^{(1)}$，$\therefore$ 能级分裂
- 将 $E_{nj}^{(1)}$ 代入能量一级修正公式，解得一组 $c_i^{(0)}$，再代回零级次波函数公式，可得相应零级波函数 $\psi_n^{(0)}$

氢原子一级斯塔克效应

斯塔克效应： 氢原子在外电场下谱线分裂的现象.

设外电场 $\vec{\mathcal{E}}$ 沿 z 轴方向.

哈密顿算符

$$\hat{H} = \hat{H}^{(0)} + \hat{H}'$$

$$\hat{H}^{(0)} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{e_s^2}{r}$$

$$\hat{H}' = e_0 \mathcal{E} r\cos\theta$$

$\hat{H}^{(0)}$ 第一激发态 $(E_n^{(0)},\ \phi_{ni})$

$$E_n^{(0)} = -\frac{\mu e_s^4}{2\hbar^2 n^2};\quad \psi_{nlm} = R_{nl}Y_{lm} \qquad n=2$$

$$(nlm) = (200),\ (210),\ (211),\ (21\bar{1})$$

$$E_2^{(0)} = -\frac{\mu e_s^4}{8\hbar^2} = -\frac{e_s^2}{8a_0},\quad a_0 = \frac{\hbar^2}{\mu e_s^2}$$

$$\phi_1 \equiv \psi_{200} = R_{20}Y_{00} = 2^{-\frac{5}{2}}\pi^{-\frac{1}{2}}a_0^{-\frac{3}{2}}\left(2-\frac{r}{a_0}\right)e^{-\frac{r}{2a_0}}$$

$$\phi_2 \equiv \psi_{210} = R_{21}Y_{10} = 2^{-\frac{5}{2}}\pi^{-\frac{1}{2}}a_0^{-\frac{3}{2}}\left(\frac{r}{a_0}\right)e^{-\frac{r}{2a_0}}\cos\theta$$

$$\phi_3 \equiv \psi_{211} = R_{21}Y_{11} = 2^{-\frac{7}{2}}\pi^{-\frac{1}{2}}a_0^{-\frac{3}{2}}\left(\frac{r}{a_0}\right)e^{-\frac{r}{2a_0}}\sin\theta e^{i\varphi}$$

$$\phi_4 \equiv \psi_{21\bar{1}} = R_{21}Y_{1\bar{1}} = 2^{-\frac{7}{2}}\pi^{-\frac{1}{2}}a_0^{-\frac{3}{2}}\left(\frac{r}{a_0}\right)e^{-\frac{r}{2a_0}}\sin\theta e^{-i\varphi}$$

微扰哈密顿的矩阵元

$$H'_{li} = \int \phi_{nl}^{*} \hat{H}' \phi_{ni} \, d\vec{\tau}$$

$$H'_{12} = \int \phi_1^{*} \hat{H}' \phi_2 \, d\vec{\tau} = H'_{21}$$

将

$$\phi_1 = \frac{a_0^{-\frac{3}{2}}}{4\sqrt{2\pi}} \left(2 - \frac{r}{a_0}\right) e^{-\frac{r}{2a_0}} \qquad \phi_2 = \frac{a_0^{-\frac{3}{2}}}{4\sqrt{2\pi}} \frac{r}{a_0} e^{-\frac{r}{2a_0}} \cos\theta$$

代入，有：

$$H'_{12} = \int \frac{a_0^{-\frac{3}{2}}}{4\sqrt{2\pi}} \left(2 - \frac{r}{a_0}\right) e^{-\frac{r}{2a_0}} e_0 \mathcal{E} r\cos\theta \frac{a^{-\frac{3}{2}}}{4\sqrt{2\pi}} \frac{r}{a_0} e^{-\frac{r}{2a_0}} \cos\theta \, d\vec{\tau}$$

$$= \frac{a^{-4}}{32\pi} e_0 \mathcal{E} \iiint r^2 \left(2 - \frac{r}{a_0}\right) e^{-\frac{r}{a_0}} \cos^2\theta \, r^2 \sin\theta \, dr d\theta d\varphi .$$

$$= \frac{a^{-4}}{32\pi} e_0 \mathcal{E} \int r^4 \left(2 - \frac{r}{a_0}\right) e^{-\frac{r}{a_0}} dr \iint \cos^2\theta \sin\theta \, d\theta d\varphi .$$

$$= \frac{a^{-4}}{32\pi} e_0 \mathcal{E} (-72 a_0^5) \frac{4}{3}\pi = -3 e_0 \mathcal{E} a_0$$

$$H'_{li} = \begin{cases} -3 e_0 \mathcal{E} a_0 & l, i = 1,2 \text{ or } 2,1 \\ 0 & \text{else} \end{cases}$$

久期方程与能级

$$H'_{Li} = \begin{cases} -3e_0\mathcal{E}a_0 & L,i = 1,2 \text{ or } 2,1 \\ 0 & else \end{cases} \qquad E_2 = E_2^{(0)} + E_2^{(1)}$$

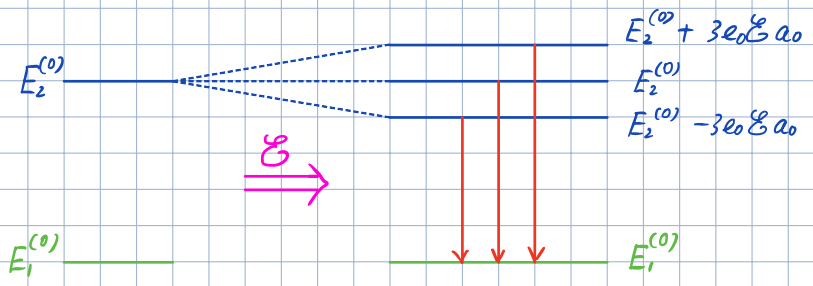
$$0 = \begin{vmatrix} H'_{11} - E_2^{(1)} & H'_{12} & H'_{13} & H'_{14} \\ H'_{21} & H'_{22} - E_2^{(1)} & H'_{23} & H'_{24} \\ H'_{31} & H'_{32} & H'_{33} - E_2^{(1)} & H'_{34} \\ H'_{41} & H'_{42} & H'_{43} & H'_{44} - E_2^{(1)} \end{vmatrix}$$

$$= \begin{vmatrix} -E_2^{(1)} & -3e_0\mathcal{E}a_0 & 0 & 0 \\ -3e_0\mathcal{E}a_0 & -E_2^{(1)} & 0 & 0 \\ 0 & 0 & -E_2^{(1)} & 0 \\ 0 & 0 & 0 & -E_2^{(1)} \end{vmatrix}$$

$$= [(E_2^{(1)})^2 - (3e_0\mathcal{E}a_0)^2](E_2^{(1)})^2$$

$$\begin{cases} E_{21}^{(1)} = 3e_0\mathcal{E}a_0 \\ E_{22}^{(1)} = -3e_0\mathcal{E}a_0 \\ E_{23}^{(1)} = E_{24}^{(1)} = 0 \end{cases} \Longrightarrow \begin{cases} E_{21} = E_2^{(0)} + 3e_0\mathcal{E}a_0 \\ E_{22} = E_2^{(0)} - 3e_0\mathcal{E}a_0 \\ E_{23} = E_{24} = E_2^{(0)} \end{cases}$$

跃迁谱线



- 四度简并的能级在外电场下分裂为3个：

  1个上移

  1个下移

  1个不变

  能级差为 $3e_0\mathcal{E}a_0$

- 零电场下的第一激发态 $\longleftrightarrow$ 基态的跃迁谱线，在外电场下分裂为3条：

  1条红移

  1条不变

  1条紫移

波函数

$$H'_{1i} = \begin{cases} 3e_0\mathcal{E}a_0 & 1,i = 1,2 \text{ or } 2,1 \\ 0 & \text{else} \end{cases}$$

$$0 = (H' - E_2^{(1)} I) c^{(0)}$$

$$= \begin{bmatrix} -E_2^{(1)} & 3e_0\mathcal{E}a_0 & 0 & 0 \\ 3e_0\mathcal{E}a_0 & -E_2^{(1)} & 0 & 0 \\ 0 & 0 & -E_2^{(1)} & 0 \\ 0 & 0 & 0 & -E_2^{(1)} \end{bmatrix} \begin{bmatrix} c_1^{(0)} \\ c_2^{(0)} \\ c_3^{(0)} \\ c_4^{(0)} \end{bmatrix}$$

$\therefore$

$$\begin{cases} 0 = E_2^{(1)} c_1^{(0)} + 3e_0\mathcal{E}a_0 \\ 0 = 3e_0\mathcal{E}a_0 + E_2^{(1)} c_2^{(0)} \\ 0 = E_2^{(1)} c_3^{(0)} \\ 0 = E_2^{(1)} c_4^{(0)} \end{cases}$$

$$\begin{cases} \psi_2^{(0)} = \sum_i c_i^{(0)} \phi_i \\ 1 = \int |\psi_2^{(0)}|^2 \, d\tau \end{cases}$$

求解：

由

$$0 = E_2^{(1)} c_1^{(0)} + 3e_0\mathcal{E}a_0$$

$$0 = 3e_0\mathcal{E}a_0 + E_2^{(1)} c_2^{(0)}$$

① when $E_2^{(1)} = 3e_0\mathcal{E}a_0$

从而：

$$\begin{cases} 0 = C_1^{(0)} + C_2^{(0)} \\ 0 = C_1^{(0)} + C_2^{(0)} \\ 0 = C_3^{(0)} \\ 0 = C_4^{(0)} \end{cases} \Longrightarrow \begin{cases} C_1^{(0)} = C \\ C_2^{(0)} = -C \\ C_3^{(0)} = 0 \\ C_4^{(0)} = 0 \end{cases}$$

$$\psi_{21}^{(0)} = C\phi_1 - C\phi_2 = (\psi_{200} - \psi_{210})/\sqrt{2}$$

② $E_{22}^{(1)} = -3e_0\mathcal{E}a_0$

$$\psi_{22}^{(0)} = (\phi_1 + \phi_2)/\sqrt{2} = (\psi_{200} + \psi_{210})/\sqrt{2}$$

③ $E_{23}^{(1)} = E_{24}^{(1)} = 0$

$$C_1^{(0)} = C_2^{(0)} = 0 \Longrightarrow C_3^{(0)} = a \quad C_4^{(0)} = b$$

$$\psi_{23}^{(0)}, \psi_{24}^{(0)} = a\phi_3 + b\phi_4 = a\psi_{211} + b\psi_{21\bar{1}}$$

说明：

1. 零级近似态 $\psi_{21}^{(0)}$ 的氢原子如同永久电偶极矩 $3e_0a_0$

2. $\psi_{21}^{(0)}$ 态中，电偶极矩与外电场同向，使总能量增加.

3. $\psi_{22}^{(0)}$ 态中，电偶极矩与外电场反向，使总能量减少.

4. $\psi_{23}^{(0)}$ $\psi_{24}^{(0)}$ 态中，电偶极矩与外电场垂直，不改变总能量.

变分法

$\hat{H}$ 的本征值方程

$$\hat{H}\psi_n = E_n\psi_n$$

将本征值由小到大排序.

$$E_0 < E_1 < E_2 < \cdots < E_n < \cdots$$

相应正交归一本征函数.

$$\psi_0,\ \psi_1,\ \psi_2,\ \cdots,\ \psi_n,\ \cdots$$

$\hat{H}$ 在任意(归一的)态中的期望值.

$$\bar{H} = \langle\phi|\hat{H}|\phi\rangle = \sum_{m,n}\langle\phi|m\rangle\langle m|\hat{H}|n\rangle\langle n|\phi\rangle \quad (\hat{H}|n\rangle = E_n|n\rangle)$$

$$= \sum_{m,n}\langle\phi|m\rangle\langle m|E_n|n\rangle\langle n|\phi\rangle = \sum_{m,n}\langle\phi|m\rangle\delta_{m,n}\langle n|\phi\rangle E_n$$

$$= \sum_n E_n\langle\phi|n\rangle\langle n|\phi\rangle$$

$$\geqslant E_0\sum_n\langle\phi|n\rangle\langle n|\phi\rangle = E_0\langle\phi|\phi\rangle = E_0$$

变分法

即在态函中引入参数，即 $\phi = \phi(\lambda)$

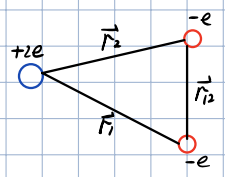
$$\bar{H}(\lambda) = \langle \phi(\lambda) | \hat{H} | \phi(\lambda) \rangle \geq E_0$$

$$\frac{d\bar{H}(\lambda)}{d\lambda} = 0 \quad \Longrightarrow \quad \bar{H}_{min} \approx E_0$$

$\bar{H}(\lambda)$ 的最小值就是 E_0 的近似值.

氦原子的基态

氦原子：1个+2e的原子核，2个-e的电子. 核位置固定.



哈密顿量

$$\hat{H} = \underbrace{-\frac{\hbar^2}{2\mu}\nabla_1^2 - \frac{\hbar^2}{2\mu}\nabla_2^2 - \frac{2e^2}{r_1} - \frac{2e^2}{r_2}}_{\hat{H}^{(0)}} + \underbrace{\frac{e^2}{r_{12}}}_{\hat{H}'}$$

$$\hat{H}^{(0)} = -\frac{\hbar^2}{2\mu}\nabla_1^2 - \frac{\hbar^2}{2\mu}\nabla_2^2 - \frac{2e^2}{r_1} - \frac{2e^2}{r_2} + \frac{e^2}{r_{12}}, \quad \hat{H}' = \frac{e^2}{r_{12}}$$

尝试波函数

$$\hat{H}^{(0)}\Psi^{(0)} = E^{(0)}\Psi^{(0)}$$

$$\Psi^{(0)}(\vec{r}_1, \vec{r}_2) = \psi_{100}(\vec{r}_1)\,\psi_{100}(\vec{r}_2) \quad \rightarrow \psi_{100}(r) = \sqrt{\frac{Z^3}{\pi a_0^3}}\, e^{-Zr/a_0}$$

$$= \sqrt{\frac{Z^3}{\pi a_0^3}}\, e^{-Zr_1/a_0} \sqrt{\frac{Z^3}{\pi a_0^3}}\, e^{-Zr_2/a_0} = \frac{Z^3}{\pi a_0^3} e^{-Z(r_1+r_2)/a_0}$$

$$\equiv \Psi$$

能量期望值

$$\bar{H} = \int \Psi^* \hat{H} \Psi \, d\vec{r}_1 \, d\vec{r}_2$$

$$= \int \frac{Z^3}{\pi a_0^3} e^{-Z(r_1+r_2)/a_0} \left(-\frac{\hbar^2\nabla_1^2}{2\mu} - \frac{\hbar^2\nabla_2^2}{2\mu} - \frac{2e^2}{r_1} - \frac{2e^2}{r_2} + \frac{e^2}{r_{12}}\right) \frac{Z^3}{\pi a_0^3} e^{-Z(r_1+r_2)/a_0} \, d\vec{r}_1 \, d\vec{r}_2$$

$$= \frac{Z^3}{\pi a_0^3} \int e^{-Z(r_1+r_2)/a_0} \left(-\frac{\hbar^2\nabla_1^2}{2\mu} - \frac{\hbar^2\nabla_2^2}{2\mu}\right) e^{-Z(r_1+r_2)/a_0} \, d\vec{r}_1 \, d\vec{r}_2$$

$$+ \frac{Z^3}{\pi a_0^3} \int e^{-2Z(r_1+r_2)/a_0} \left(-\frac{2e^2}{r_1} - \frac{2e^2}{r_2}\right) d\vec{r}_1 \, d\vec{r}_2 + \frac{Z^3}{\pi a_0^3} \int e^{-2Z(r_1+r_2)/a_0} \frac{e^2}{r_{12}} \, d\vec{r}_1 \, d\vec{r}_2$$

$$= \frac{e_s^2 Z^2}{a_0} - \frac{4e_s^2 Z}{a_0} + \frac{5e_s^2 Z}{8a_0} = \frac{e_s^2}{a_0}\left(Z^2 - \frac{27}{8}Z\right)$$

以 Z 为参数进行变分：

$$\frac{d\bar{H}}{dZ} = \frac{e_s^2}{a_0}\left(2Z - \frac{27}{8}\right) = 0$$

$$Z = \frac{27}{16} \approx 1.69$$

$$E_0 \approx \bar{H}\big|_{Z=1.69} = -2.85\frac{e_s^2}{a_0}$$

$$\Psi = \frac{27^3}{16^3 \pi a_0^3} e^{-\frac{27}{16a_0}(r_1+r_2)}$$

说明：

① $Z<2$，由于 2 个电子相互屏蔽，核的有效电荷不再为 $2e$.

② 实验测得：$E_0 = -2.90 e_s^2/a_0$，表明：变分法结果比一级微扰法好，原因是 $\bar{H}' = e_s^2/r_{12}$ 接近于 $\hat{H}$ 中的 $2e_s^2/r_1 + 2e_s^2/r_2$ 不满足微扰法使用情况.

③ 变分法的改良：选取更适当的尝试波函数，增加变分参数.

时间的微扰

显含时间的哈密顿量 $\hat{H}(t)$ 分为：

- $\hat{H}^{(0)}$ 不含时间主要部分
- $\hat{H}'(t)$ 含时间微扰部分

$$\hat{H}^{(0)}\phi_n = \varepsilon_n \phi_n$$

Schrodinger eq 的解的时间演化

$$\Psi = \sum_n a_n \Phi_n$$

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi \qquad i\hbar \frac{\partial \Phi_n}{\partial t} = \hat{H}^{(0)}\Phi_n$$

$$\hat{H}^{(0)}\phi_n = \varepsilon_n \phi_n \qquad \Phi_n = \phi_n e^{-i\varepsilon_n t/\hbar}$$

$$Left = i\hbar \frac{\partial}{\partial t}\sum_n a_n \Phi_n = i\hbar \sum_n \frac{\partial}{\partial t} a_n \Phi_n = i\hbar \sum_n \left(\frac{da_n}{dt}\phi_n + a_n \frac{\partial \Phi_n}{\partial t}\right)$$

$$= i\hbar \sum_n \frac{da_n}{dt}\Phi_n + \underbrace{\sum_n a_n \hat{H}^{(0)}\Phi_n}_{\text{消去}}$$

$$Right = \hat{H}\sum_n a_n \Phi_n = \left(\hat{H}^{(0)} + \hat{H}'(t)\right)\sum_n a_n \Phi_n = \underbrace{\sum_n a_n \hat{H}^{(0)}\Phi_n}_{\text{消去}} + \sum_n a_n \hat{H}'\Phi_n$$

$$\Longrightarrow \quad \int \Phi_m^* \, i\hbar \sum_n \frac{da_n}{dt}\Phi_n \, d\tau = \int \Phi_m^* \sum_n a_n \hat{H}' \Phi_n \, d\tau$$

$$Left = i\hbar \sum_n \frac{da_n}{dt}\int \Phi_m^* \Phi_n \, d\tau = i\hbar \sum_n \frac{da_n}{dt}\delta_{mn} = i\hbar \frac{da_m}{dt}$$

$$Right = \sum_n a_n \int \Phi_m^* \hat{H}' \Phi_n \, d\tau = \sum_n a_n \int \phi_m^* e^{i\varepsilon_m t/\hbar} \hat{H}' \left(\phi_n e^{i\varepsilon_n t/\hbar}\right) d\tau$$

$$= \sum_n a_n e^{i(\varepsilon_m - \varepsilon_n)t/\hbar} \int \phi_m^* \hat{H}' \phi_n \, d\tau = \sum_n a_n e^{i(\varepsilon_m - \varepsilon_n)t/\hbar} H'_{mn}$$

$\omega_{mn}=\frac{\varepsilon_m-\varepsilon_n}{\hbar}$ $\sum_n a_n e^{i\omega_{mn}t} H'_{mn}$

综上，有：

$$\Psi = \sum_n a_n \phi_n e^{i\varepsilon_n t/\hbar}$$

$$i\hbar \frac{da_m}{dt} = \sum_n a_n e^{i\omega_{mn}t} H'_{mn}$$

其中，$\omega_{mn} = \frac{\varepsilon_m - \varepsilon_n}{\hbar}$ 为体系由 ε_n 能级跃迁至 ε_m 能级的玻尔频率

初始条件：$t=0$. 体系处于 $\hat{H}^{(0)}$ 本征态 Φ_k，且微扰开始.

$$a_n|_{t=0} = \delta_{nk}$$

Ψ（即 a_m）的一级近似解.

$$a_n(t) = a_n(0) + a_n'(0)t + \cdots \approx a_n(0) = \delta_{nk}.$$

$$i\hbar \frac{da_m}{dt} = \sum_n a_n e^{i\omega_{mn}t} H'_{mn} = \sum_n \delta_{nk} e^{i\omega_{mn}t} H'_{mn}$$

$$= e^{i\omega_{mn}t} H'_{mk}$$

$$a_m = \frac{1}{i\hbar}\int_0^t e^{i\omega_{mn}t'} H'_{mk}\, dt'$$

微扰作用下，体系由 Φ_k 跃迁至 Φ_m 的概率为：

$$W_{k\to m} = |a_m|^2$$

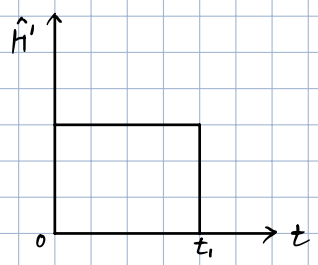
跃迁概率

计算2种不同情况下的 $a_m(t)$ 与 $W_{k\to m}$

1. $\hat{H}'$ 在 $0\le t\le t_1$ 为恒定的微扰

ϕ_k $\xrightarrow{H'作用下}$ 一系列连续 ϕ_m

$(t=0)$ ε_m 连续. $(t>t_1)$



设能量在 $\varepsilon_m\sim\varepsilon_m+d\varepsilon_m$ 间态数目为 $\rho(m)d\varepsilon_m$

末态的态密度.

1. $t\to\infty$ 时，a_m 的一级近似解为：

$$a_m^{(1)}=\frac{1}{i\hbar}\int_0^t e^{i\omega_{mk}t'}H'_{mk}dt'=\frac{H'_{mk}}{i\hbar}\frac{e^{i\omega_{mk}t'}}{i\omega_{mk}}\Big|_0^t=-\frac{H'_{mk}}{\hbar}\frac{e^{i\omega_{mk}t}-1}{\omega_{mk}}$$

$$|a_m^{(1)}|^2=\left|\frac{H'_{mk}}{\hbar}\frac{e^{i\omega_{mk}t}-1}{\omega_{mk}}\right|^2=\frac{|H'_{mk}|^2}{\hbar^2\omega_{mk}^2}4\sin^2\frac{\omega_{mk}t}{2}=\frac{|H'_{mk}|^2}{\hbar^2}2\pi t\frac{\sin^2(\omega_{mk}\frac{t}{2})}{\pi\omega_{mk}^2\frac{t}{2}}$$

$$=\frac{|H'_{mk}|^2}{\hbar^2}2\pi t\,\delta(\omega_{mk})$$

$t\to\infty$

$$\lim_{t\to\infty}\frac{\sin^2(xt)}{\pi x^2t}=\delta(x)$$

2. 跃迁概率

为各种跃迁概率的和.

$$W=\int_{-\infty}^{\infty}|a_m|^2\rho(m)\,d\underbrace{\varepsilon_m}_{\varepsilon_m=\varepsilon_k+\hbar\omega_{mk}}=\int_{-\infty}^{\infty}\frac{2\pi t|H'_{mk}|^2}{\hbar^2}\delta(\omega_{mk})\rho(m)\hbar\,d\omega_{mk}$$

$$=\frac{2\pi t|H'_{mk}|^2}{\hbar}\rho(m)$$

费米黄金规则：

单位时间跃迁概率：

$$w=\frac{W}{t}=\frac{2\pi}{\hbar}|H'_{mk}|^2\rho(m)$$

3. 态密度.

$\rho(m)$ 形式决定于系统末态具体情况.

例如：自由粒子

末态为自由粒子动量的本征函数，采用箱归一化：

$$\varphi_m(r)=L^{-\frac{3}{2}}\exp\left(\frac{i}{\hbar}\vec{p}\cdot\vec{r}\right)$$

动量本征值：

$$p_x=\frac{2\pi\hbar n_x}{L}\qquad p_y=\frac{2\pi\hbar n_y}{L}\qquad p_z=\frac{2\pi\hbar n_z}{L}$$

有一组 (n_x,n_y,n_z) 确定一个态.

∴ 动量在 $p_x\to p_x+dp_x$，$p_y\to p_x+dp_y$，$p_z\to p_z+dp_z$ 范围态数目为

$$\left(\frac{L}{2\pi\hbar}\right)^3dp_x\,dp_y\,dp_z=\left(\frac{L}{2\pi\hbar}\right)^3p^2dp\sin\theta\,d\theta d\varphi$$

能量

$$\varepsilon_m = \frac{p^2}{2m} \qquad d\varepsilon_m = \frac{p}{m}dp$$

$$\rho(m)\,d\varepsilon_m = \left(\frac{L}{2\pi\hbar}\right)^3 p^2 dp \sin\theta\, d\theta d\varphi = \left(\frac{L}{2\pi\hbar}\right)^3 mp\sin\theta\, d\theta d\varphi$$